6

Cold plasma waves in a magnetized plasma

6.1 Overview

Chapter 4 showed that finite temperature is responsible for the lowest order dispersive terms in both electron plasma waves (dispersion $\omega^2 = \omega_p^2 + 3k^2\kappa T_e/m_e$) and ion acoustic waves (dispersion $\omega^2 = k^2 c_s^2/(1+k^2\lambda_{De}^2)$). Furthermore, finite temperature was shown in Chapter 5 to be essential to Landau damping and instability.

Chapter 4 also contained a derivation of the electromagnetic plasma wave (dispersion $\omega^2 = \omega_{pe}^2 + k^2c^2$) and of the inertial Alfvén wave (dispersion $\omega^2 = k_z^2 v_A^2/(1+k_x^2c^2/\omega_{pe}^2)$), both of which had no dependence on temperature. To distinguish waves that depend on temperature from waves that do not, the terminology "cold plasma wave" and "hot plasma wave" is used. A cold plasma wave is a wave having a temperature-independent dispersion relation so that the temperature could be set to zero without changing the wave, whereas a hot plasma wave has a temperature-dependent dispersion relation. Thus, hot and cold do not refer to a "temperature" of the wave, but rather to the wave's dependence or lack thereof on plasma temperature. Generally speaking, cold plasma waves are just the consequence of a large number of particles having identical Hamiltonian–Lagrangian dynamics whereas hot plasma waves involve different groups of particles having different dynamics because they have different initial velocities. Thus, hot plasma waves involve statistical mechanical or thermodynamic considerations. The general theory of cold plasma waves in a uniform magnetized plasma is presented in this chapter and hot plasma waves will be discussed in later chapters.

6.2 Redundancy of Poisson's equation in electromagnetic mode analysis

When electrostatic waves were examined in Chapter 4 it was seen that the plasma response to the wave electric field could be expressed as a sum of susceptibilities, where the susceptibility of each species was proportional to the density perturbation of that species. Combining the susceptibilities with Poisson's equation gave a dispersion relation. However, because electric fields can also be generated inductively, electrostatic waves are not the only type of wave. Inductive electric fields result from time-dependent currents, i.e., from charged particle acceleration, and do not involve density perturbations. As an example, the electromagnetic plasma wave involved inductive rather than electrostatic electric fields. The inertial Alfvén wave involved inductive electric fields in the parallel direction and electrostatic electric fields in the perpendicular direction.

One might expect that a procedure analogous to the previous derivation of electrostatic susceptibilities could be used to derive inductive "susceptibilities," which would then be used to construct dispersion relations for inductive modes. It turns out that such a procedure not only gives dispersion relations for inductive modes, but also includes the electrostatic modes. Thus, it turns out to be unnecessary to analyze electrostatic modes separately. The main reason for investigating electrostatic modes separately as done earlier is pedagogical – it is easier to understand a simpler system. To see why electrostatic modes are automatically included in an electromagnetic analysis, consider the interrelationship between Poisson's equation, Ampère's law, and a charge-weighted summation of the two-fluid continuity equation,

$$\nabla\cdot\mathbf{E} = -\frac{1}{\varepsilon_0}\sum_\sigma n_\sigma q_\sigma, \tag{6.1}$$

$$\nabla\times\mathbf{B} = \mu_0\mathbf{J} + \varepsilon_0\mu_0\frac{\partial\mathbf{E}}{\partial t}, \tag{6.2}$$

$$\sum_\sigma q_\sigma\left[\frac{\partial n_\sigma}{\partial t} + \nabla\cdot(n_\sigma\mathbf{v}_\sigma)\right] = \left[\frac{\partial}{\partial t}\sum_\sigma n_\sigma q_\sigma\right] + \nabla\cdot\mathbf{J} = 0. \tag{6.3}$$

The divergence of Eq. (6.2) gives

$$\nabla\cdot\mathbf{J} + \varepsilon_0\frac{\partial}{\partial t}\nabla\cdot\mathbf{E} = 0 \tag{6.4}$$

and substituting Eq. (6.3) gives

$$\frac{\partial}{\partial t}\left[-\sum_\sigma n_\sigma q_\sigma + \varepsilon_0\nabla\cdot\mathbf{E}\right] = 0, \tag{6.5}$$

which is just the time derivative of Poisson's equation. Integrating Eq. (6.5) shows

$$-\sum_\sigma n_\sigma q_\sigma + \varepsilon_0\nabla\cdot\mathbf{E} = const. \tag{6.6}$$

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Poisson's equation, Eq. (6.1), thus provides an *initial condition*, which fixes the value of the constant in Eq. (6.6). Since all small-amplitude perturbations are assumed to have the phase dependence $\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t)$ and therefore behave as a single Fourier mode, the $\partial/\partial t$ operator in Eq. (6.5) is replaced by $-\mathrm{i}\omega$, in which case the constant in Eq. (6.6) is automatically set to zero, making a separate consideration of Poisson's equation redundant. In summary, the Fourier-transformed Ampère's law effectively embeds Poisson's equation and so a discussion of waves based solely on currents describes inductive and electrostatic modes as well as modes, such as the inertial Alfvén wave, that involve a mixture of inductive and electrostatic electric fields.

6.3 Dielectric tensor

Section 3.8 showed that a single particle immersed in a constant, uniform equilibrium magnetic field $\mathbf{B}=B_0\hat{z}$ and subject to a *small-amplitude* wave with electric field $\sim \exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t)$ has the velocity

$$\tilde{\mathbf{v}}_\sigma = \frac{\mathrm{i}q_\sigma}{\omega m_\sigma}\left[\tilde{E}_z\hat{z} + \frac{\tilde{\mathbf{E}}_\perp}{1-\omega_{c\sigma}^2/\omega^2} - \frac{\mathrm{i}\omega_{c\sigma}}{\omega}\frac{\hat{z}\times\tilde{\mathbf{E}}}{1-\omega_{c\sigma}^2/\omega^2}\right]\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t}. \tag{6.7}$$

The tilde ˜ denotes a small-amplitude oscillatory quantity with space-time dependence $\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t)$; this phase factor may or may not be explicitly written, but should always be understood to exist for a tilde-denoted quantity.

The three terms in Eq. (6.7) are respectively:

1. *The parallel quiver velocity*. This quiver velocity is the same as the quiver velocity of an unmagnetized particle, but is restricted to parallel motion. Because the magnetic force $q(\mathbf{v}\times\mathbf{B})$ vanishes for motion along the magnetic field, motion parallel to $\mathbf{B}$ in a magnetized plasma is identical to motion in an unmagnetized plasma.
2. *The generalized polarization drift*. This motion has a resonance at the cyclotron frequency but at low frequencies such that $\omega \ll \omega_{c\sigma}$, it reduces to the polarization drift $\mathbf{v}_{p\sigma} = m_\sigma\dot{\mathbf{E}}_\perp/q_\sigma B^2$ derived in Chapter 3.
3. *The generalized* $\mathbf{E}\times\mathbf{B}$ *drift*. This also has a resonance at the cyclotron frequency and for $\omega \ll \omega_{c\sigma}$ reduces to the drift $\mathbf{v}_E = \mathbf{E}\times\mathbf{B}/B^2$ derived in Chapter 3.

The particle velocities given by Eq. (6.7) produce a plasma current density

$$\begin{aligned}\tilde{\mathbf{J}} &= \sum_\sigma n_{0\sigma}q_\sigma\tilde{\mathbf{v}}_\sigma \\ &= \mathrm{i}\varepsilon_0\sum_\sigma\frac{\omega_{p\sigma}^2}{\omega}\left[\tilde{E}_z\hat{z} + \frac{\tilde{\mathbf{E}}_\perp}{1-\omega_{c\sigma}^2/\omega^2} - \frac{\mathrm{i}\omega_{c\sigma}}{\omega}\frac{\hat{z}\times\tilde{\mathbf{E}}}{1-\omega_{c\sigma}^2/\omega^2}\right]\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}-\mathrm{i}\omega t}.\end{aligned} \tag{6.8}$$

If these plasma currents are written out explicitly, then Ampère's law has the form

$$\begin{aligned}\nabla \times \tilde{\mathbf{B}} &= \mu_0 \tilde{\mathbf{J}} + \mu_0 \varepsilon_0 \frac{\partial \tilde{\mathbf{E}}}{\partial t} \\ &= \mu_0 \left(\mathrm{i}\varepsilon_0 \sum_\sigma \frac{\omega_{p\sigma}^2}{\omega} \left[\tilde{E}_z \hat{z} + \frac{\tilde{\mathbf{E}}_\perp}{1-\omega_{c\sigma}^2/\omega^2} - \frac{\mathrm{i}\omega_{c\sigma}}{\omega} \frac{\hat{z} \times \tilde{\mathbf{E}}}{1-\omega_{c\sigma}^2/\omega^2} \right] - \mathrm{i}\omega\varepsilon_0 \tilde{\mathbf{E}} \right),\end{aligned} \tag{6.9}$$

where a factor $\exp\left(\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t\right)$ is implicit.

The cold plasma wave equation is established by combining Ampère's and Faraday's laws in a manner similar to the method used for vacuum electromagnetic waves. However, before doing so, it is useful to define the dielectric tensor $\overleftrightarrow{\mathbf{K}}$. This tensor contains the information in the right-hand side of Eq. (6.9) so that this equation is written as

$$\nabla \times \tilde{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\overleftrightarrow{\mathbf{K}} \cdot \tilde{\mathbf{E}} \right), \tag{6.10}$$

where

$$\begin{aligned}\overleftrightarrow{\mathbf{K}} \cdot \tilde{\mathbf{E}} &= \tilde{\mathbf{E}} - \frac{\omega_{p\sigma}^2}{\omega^2} \left[\tilde{E}_z \hat{z} + \frac{\tilde{\mathbf{E}}_\perp}{1-\omega_{c\sigma}^2/\omega^2} - \frac{\mathrm{i}\omega_{c\sigma}}{\omega} \frac{\hat{z} \times \tilde{\mathbf{E}}}{1-\omega_{c\sigma}^2/\omega^2} \right] \\ &= \begin{bmatrix} S & -\mathrm{i}D & 0 \\ \mathrm{i}D & S & 0 \\ 0 & 0 & P \end{bmatrix} \cdot \tilde{\mathbf{E}}\end{aligned} \tag{6.11}$$

and the elements of the dielectric tensor are

$$S = 1 - \sum_{\sigma=i,e} \frac{\omega_{p\sigma}^2}{\omega^2 - \omega_{c\sigma}^2}, \qquad D = \sum_{\sigma=i,e} \frac{\omega_{c\sigma}}{\omega} \frac{\omega_{p\sigma}^2}{\omega^2 - \omega_{c\sigma}^2}, \quad P = 1 - \sum_{\sigma=i,e} \frac{\omega_{p\sigma}^2}{\omega^2}. \tag{6.12}$$

The nomenclature S, D, P for the matrix elements was introduced by Stix (1962) and is a mnemonic for "Sum," "Difference," and "Parallel." The reasoning behind "Sum" and "Difference" will become apparent later, but for now it is clear that the P element corresponds to the cold plasma limit of the parallel dielectric, i.e., $P = 1 + \chi_i + \chi_e$, where $\chi_\sigma = -\omega_{p\sigma}^2/\omega^2$. This is the cold limit of the unmagnetized dielectric because behavior involving parallel motions in a magnetized plasma is identical to that in an unmagnetized plasma. In the limit of no plasma, $\overleftrightarrow{\mathbf{K}}$ becomes the unit tensor and describes the effect of the vacuum displacement current only.

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This definition of the dielectric tensor means that Maxwell's equations, the Lorentz equation, and the plasma currents can now be summarized in just two coupled equations, namely

$$\nabla \times \mathbf{B} = \frac{1}{c^2}\frac{\partial}{\partial t}\left(\overleftrightarrow{\mathbf{K}} \cdot \mathbf{E}\right) \tag{6.13}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{6.14}$$

where, for clarity, the tildes have now been omitted and it is to be understood that **E** and **B** refer to the wave fields. The cold plasma wave equation is obtained by taking the curl of Eq. (6.14) and then substituting for $\nabla \times \mathbf{B}$ using (6.13) to obtain

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\left(\overleftrightarrow{\mathbf{K}} \cdot \mathbf{E}\right). \tag{6.15}$$

Since a phase dependence $\exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t)$ is assumed, this can be written in algebraic form as

$$\mathbf{k} \times (\mathbf{k} \times \mathbf{E}) = -\frac{\omega^2}{c^2}\overleftrightarrow{\mathbf{K}} \cdot \mathbf{E}. \tag{6.16}$$

It is now convenient to define the refractive index $\mathbf{n} = c\mathbf{k}/\omega$, a renormalization of the wavevector **k** arranged so that light waves have a refractive index of unity. Using this definition Eq. (6.16) becomes

$$\mathbf{nn} \cdot \mathbf{E} - n^2\mathbf{E} + \overleftrightarrow{\mathbf{K}} \cdot \mathbf{E} = 0, \tag{6.17}$$

which is essentially a set of three homogeneous equations in the three components of **E**.

The refractive index $\mathbf{n} = c\mathbf{k}/\omega$ can be decomposed into parallel and perpendicular components relative to the equilibrium magnetic field $\mathbf{B} = B_0\hat{z}$. For convenience, the x axis of the coordinate system is defined to lie along the perpendicular component of **n** so that $n_y = 0$ by assumption. This simplification is possible for a spatially uniform equilibrium only; if the plasma is non-uniform in the $x-y$ plane, there can be a real distinction between x and y direction propagation and the refractive index in the y direction cannot be simply defined away by choice of coordinate system.

To set the stage for obtaining a dispersion relation, Eq. (6.17) is written in matrix form as

$$\begin{bmatrix} S - n_z^2 & -\mathrm{i}D & n_x n_z \\ \mathrm{i}D & S - n^2 & 0 \\ n_x n_z & 0 & P - n_x^2 \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0. \tag{6.18}$$

It is now useful to introduce a spherical coordinate system in k-space (or equivalently refractive index space) with $\hat{z}$ defining the axis and θ the polar angle.

Thus, the Cartesian components of the refractive index are related to the spherical components by

$$
\begin{aligned}
n_x &= n\sin\theta \\
n_z &= n\cos\theta \\
n^2 &= n_x^2 + n_z^2
\end{aligned}
\tag{6.19}
$$

and so Eq. (6.18) becomes

$$
\begin{bmatrix} S - n^2\cos^2\theta & -\mathrm{i}D & n^2\sin\theta\cos\theta \\ \mathrm{i}D & S - n^2 & 0 \\ n^2\sin\theta\cos\theta & 0 & P - n^2\sin^2\theta \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0. \tag{6.20}
$$

6.3.1 Mode behavior at $\theta = 0$

Non-trivial solutions to the set of three coupled equations for E_x, E_y, E_z prescribed by Eq. (6.20) exist only if the determinant of the matrix vanishes. For arbitrary values of θ, this determinant is complicated. Rather than examining the arbitrary-θ determinant immediately, two simpler limiting cases will first be considered, namely the situations where $\theta = 0$ (i.e., $\mathbf{k} \parallel \mathbf{B}_0$) and $\theta = \pi/2$ (i.e., $\mathbf{k} \perp \mathbf{B}_0$). These special cases are simpler than the general case because the off-diagonal matrix elements $n^2\sin\theta\cos\theta$ vanish for both $\theta = 0$ and $\theta = \pi/2$.

When $\theta = 0$ Eq. (6.20) becomes

$$
\begin{bmatrix} S - n^2 & -\mathrm{i}D & 0 \\ \mathrm{i}D & S - n^2 & 0 \\ 0 & 0 & P \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0. \tag{6.21}
$$

The determinant of this system is

$$
\left[\left(S - n^2\right)^2 - D^2\right] P = 0, \tag{6.22}
$$

which has roots

$$
P = 0 \tag{6.23}
$$

and

$$
n^2 - S = \pm D. \tag{6.24}
$$

Equation (6.24) may be rearranged in the form

$$
n^2 = R, \quad n^2 = L, \tag{6.25}
$$

where

$$
R = S + D, \quad L = S - D \tag{6.26}
$$

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have the mnemonics "right" and "left." The rationale behind the nomenclature "S(um)" and "D(ifference)" now becomes apparent since

$$S = \frac{R+L}{2}, \quad D = \frac{R-L}{2}. \tag{6.27}$$

What does all this algebra mean? Equation (6.25) states that for $\theta = 0$ the dispersion relation has two distinct roots, each corresponding to a natural mode (or characteristic wave) constituting a self-consistent solution to the Maxwell–Lorentz system. The definitions in Eqs. (6.12) and (6.26) show that

$$R = 1 - \sum_\sigma \frac{\omega_{p\sigma}^2}{\omega(\omega + \omega_{c\sigma})}, \quad L = 1 - \sum_\sigma \frac{\omega_{p\sigma}^2}{\omega(\omega - \omega_{c\sigma})} \tag{6.28}$$

so that R diverges when $\omega = -\omega_{c\sigma}$ whereas L diverges when $\omega = \omega_{c\sigma}$. Since $\omega_{c\sigma} = q_\sigma B/m_\sigma$, the ion cyclotron frequency is positive and the electron cyclotron frequency is negative. Hence, R diverges at the electron cyclotron frequency, whereas L diverges at the ion cyclotron frequency. When $\omega \to \infty$, both $R,\ L \to 1$. In the limit $\omega \to 0$, evaluation of $R,\ L$ must be done very carefully, since

$$\begin{aligned} \frac{\omega_{p\sigma}^2}{\omega_{c\sigma}} &= \frac{n_\sigma q_\sigma^2}{\varepsilon_0 m_\sigma} \frac{m_\sigma}{q_\sigma B} \\ &= \frac{n_\sigma q_\sigma}{\varepsilon_0 B} \end{aligned} \tag{6.29}$$

so

$$\frac{\omega_{pi}^2}{\omega_{ci}} = -\frac{\omega_{pe}^2}{\omega_{ce}}. \tag{6.30}$$

Thus

$$\begin{aligned} \lim_{\omega \to 0} R, L &= 1 - \frac{1}{\omega}\left[\frac{\omega_{pi}^2}{(\omega \pm \omega_{ci})} + \frac{\omega_{pe}^2}{(\omega \pm \omega_{ce})}\right] \\ &= 1 - \frac{\omega_{pi}^2 + \omega_{pe}^2}{\omega_{ci}\omega_{ce}} \\ &\simeq 1 - \frac{n_e q_e^2}{\varepsilon_0 m_e} \frac{m_i}{q_i B} \frac{m_e}{q_e B} \\ &= 1 + \frac{\omega_{pi}^2}{\omega_{ci}^2} \\ &= 1 + \frac{c^2}{v_A^2}, \end{aligned} \tag{6.31}$$

where $v_A^2 = B^2/\mu_0\rho$ is the Alfvén velocity. Thus, at low frequency, both R and L are related to Alfvén modes. The $n^2 = L$ mode is the slow mode (larger k) and

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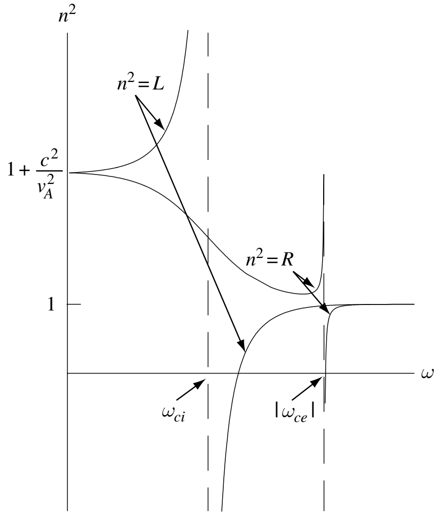

Fig. 6.1 Propagation parallel to the magnetic field.

the $n^2 = R$ mode is the fast mode (smaller k). Figure 6.1 shows the frequency dependence of the $n^2 = R, L$ modes.

Having determined the eigenvalues for $\theta = 0$, the associated eigenvectors can now be found. These are obtained by substituting the eigenvalue back into the original set of equations; for example, substitution of $n^2 = R$ into Eq. (6.21) gives

$$\begin{bmatrix} -D & -\mathrm{i}D & 0 \\ \mathrm{i}D & -D & 0 \\ 0 & 0 & P \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0, \tag{6.32}$$

so that the eigenvector associated with $n^2 = R$ is

$$\frac{E_x}{E_y} = -\mathrm{i}, \quad \text{for eigenvalue } n^2 = R. \tag{6.33}$$

The implication of this eigenvector can be seen by considering the root $n = +\sqrt{R}$ so that the electric field in the plane orthogonal to $\hat{z}$ has the form

$$\begin{aligned} \mathbf{E}_\perp &= \mathrm{Re}\left\{E_\perp\left(\hat{x} + \mathrm{i}\hat{y}\right)\exp\left(\mathrm{i}k_z z - \mathrm{i}\omega t\right)\right\} \\ &= |E_\perp|\left\{\hat{x}\cos(k_z z - \omega t + \delta) - \hat{y}\sin(k_z z - \omega t + \delta)\right\}, \end{aligned} \tag{6.34}$$

where $E_\perp = |E_\perp|\mathrm{e}^{\mathrm{i}\delta}$. This is a *right-hand* circularly polarized wave propagating in the positive z direction; hence the nomenclature R. Similarly, the $n^2 = L$ root gives a left-hand circularly polarized wave. Linearly polarized waves may be

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constructed from appropriate sums and differences of these left- and right-hand circularly polarized waves.

In summary, two distinct modes exist when the wavevector happens to be exactly parallel to the magnetic field ($\theta = 0$): a right-hand circularly polarized wave with dispersion $n^2 = R$ with $n \to \infty$ at the electron cyclotron resonance and a left-hand circularly polarized mode with dispersion $n^2 = L$ with $n \to \infty$ at the ion cyclotron resonance. Since ion cyclotron motion is left-handed (mnemonic "Lion") it is reasonable that a left-hand circularly polarized wave resonates with ions, and vice versa for electrons. At low frequencies, these modes become Alfvén modes with dispersion $n_z^2 = 1 + c^2/v_A^2$ for $\theta = 0$. In the Chapter 4 discussion of Alfvén modes the dispersions of both compressional and shear modes were found to reduce to $c^2k_z^2/\omega^2 = n_z^2 = c^2/v_A^2$ for $\theta = 0$. One may ask why a "1" term did not appear in the Chapter 4 dispersion relations. The answer is that the "1" term comes from displacement current, a quantity neglected in the Chapter 4 derivations. The displacement current term shows that if the plasma density is so low (or the magnetic field is so high) that v_A becomes larger than c, then Alfvén modes become ordinary vacuum electromagnetic waves propagating at nearly the speed of light. In order for a plasma to demonstrate significant Alfvénic (i.e., MHD) behavior it must satisfy $B/\sqrt{\mu_0\rho} \ll c$ or equivalently have $\omega_{ci} \ll \omega_{pi}$.

6.3.2 Cutoffs and resonances

The general situation where $n^2 \to \infty$ is called a *resonance* and corresponds to the wavelength going to zero. Any slight dissipative effect in this situation will cause large wave damping. This is because if the wavelength becomes infinitesimal and the fractional attenuation per wavelength is constant, there will be a near-infinite number of wavelengths and the wave amplitude is reduced by the same fraction for each of these. Figure 6.1 also shows it is possible to have a situation where $n^2 = 0$. The general situation where $n^2 = 0$ is called a *cutoff* and corresponds to wave reflection, since n changes from being pure real to pure imaginary. If the plasma is non-uniform, it is possible for layers to exist in the plasma where either $n^2 \to \infty$ or $n^2 = 0$; these are called resonance or cutoff layers. Typically, if a wave intercepts a resonance layer it is absorbed, whereas if it intercepts a cutoff layer it is reflected.

6.3.3 Mode behavior at $\theta = \pi/2$

When $\theta = \pi/2$ Eq. (6.20) becomes

$$\begin{bmatrix} S & -\mathrm{i}D & 0 \\ \mathrm{i}D & S - n^2 & 0 \\ 0 & 0 & P - n^2 \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0 \tag{6.35}$$

and again two distinct modes appear. The first mode has as its eigenvector the condition $E_z \neq 0$. The associated eigenvalue equation is $P - n^2 = 0$ or

$$\omega^2 = k^2c^2 + \sum_\sigma \omega_{p\sigma}^2, \tag{6.36}$$

which is just the dispersion for an electromagnetic plasma wave in an unmagnetized plasma. This is in accordance with the prediction that modes involving particle motion strictly parallel to the magnetic field are unaffected by the magnetic field. This mode is called the *ordinary* mode because it is unaffected by the magnetic field.

The second mode involves both E_x and E_y and has the eigenvalue equation $S(S-n^2) - D^2 = 0$, which gives the dispersion relation

$$n^2 = \frac{S^2 - D^2}{S} = 2\frac{RL}{R+L}. \tag{6.37}$$

Cutoffs occur here when either $R = 0$ or $L = 0$ and a resonance occurs when $S = 0$. Since this mode depends on the magnetic field, it is called the *extraordinary* mode. The $S = 0$ resonance is called a hybrid resonance because it depends on a hybrid of $\omega_{c\sigma}^2$ and $\omega_{p\sigma}^2$ terms (the $\omega_{c\sigma}^2$ terms depend on single-particle physics, whereas the $\omega_{p\sigma}^2$ terms depend on collective motion physics). Because S is quadratic in ω^2, the equation $S = 0$ has two distinct roots and these are found by explicitly writing

$$S = 1 - \frac{\omega_{pi}^2}{\omega^2 - \omega_{ci}^2} - \frac{\omega_{pe}^2}{\omega^2 - \omega_{ce}^2} = 0. \tag{6.38}$$

A plot of this expression shows the two roots are well separated. The large root may be found by assuming $\omega \sim \mathcal{O}(\omega_{ce})$, in which case the ion term becomes insignificant. Dropping the ion term shows the large root of S is simply

$$\omega_{uh}^2 = \omega_{pe}^2 + \omega_{ce}^2, \tag{6.39}$$

which is called the *upper hybrid frequency*. The small root may be found by assuming that $\omega^2 \ll \omega_{ce}^2$, which gives the *lower hybrid frequency*

$$\omega_{lh}^2 = \omega_{ci}^2 + \frac{\omega_{pi}^2}{1 + \dfrac{\omega_{pe}^2}{\omega_{ce}^2}}. \tag{6.40}$$

6.3.4 Very-low-frequency modes where θ is arbitrary

Equation (6.31) shows that for $\omega \ll \omega_{ci}$

$$\begin{aligned} S \simeq R \simeq L &\simeq 1 + c^2/v_A^2 \\ D &\simeq 0 \end{aligned} \tag{6.41}$$

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so the cold plasma dispersion simplifies to

$$\begin{bmatrix} S-n_z^2 & 0 & n_x n_z \\ 0 & S-n^2 & 0 \\ n_x n_z & 0 & P-n_x^2 \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = 0. \tag{6.42}$$

Because $D=0$ the determinant factors into two modes, one where

$$\left(S-n^2\right) E_y = 0 \tag{6.43}$$

and the other where

$$\begin{bmatrix} S-n_z^2 & n_x n_z \\ n_x n_z & P-n_x^2 \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_z \end{bmatrix} = 0. \tag{6.44}$$

The former gives the dispersion relation

$$n^2 = S \tag{6.45}$$

with $E_y \neq 0$ as the eigenvector. This mode is the fast or compressional mode since, in the limit where the displacement current can be neglected, Eq. (6.45) becomes $\omega^2 = k^2 v_A^2$. The latter mode involves finite E_x and E_z and has the dispersion

$$n_x^2 = \frac{P}{S}\left(S-n_z^2\right), \tag{6.46}$$

which is the inertial Alfvén wave $\omega^2 = k_z^2 v_A^2 / \left(1+k_x^2 c^2/\omega_{pe}^2\right)$ in the limit where the displacement current can be neglected.

6.3.5 Modes where ω and θ are arbitrary

The $\theta = 0,\ \pi/2$ limiting behaviors and the low-frequency Alfvén modes gave a useful introduction to the cold plasma modes and, in particular, showed how modes can be subject to cutoffs or resonances. We now evaluate the determinant of the matrix in Eq. (6.20) for arbitrary θ and arbitrary ω; after some algebra this determinant can be written as

$$An^4 - Bn^2 + C = 0, \tag{6.47}$$

where

$$\begin{aligned} A &= S\sin^2\theta + P\cos^2\theta \\ B &= (S^2-D^2)\sin^2\theta + PS(1+\cos^2\theta) \\ C &= P(S^2-D^2) = PRL. \end{aligned} \tag{6.48}$$

Equation (6.47) is quadratic in n^2 and has the two roots

$$n^2 = \frac{B \pm \sqrt{B^2 - 4AC}}{2A}. \tag{6.49}$$

Thus, the two distinct modes in the special cases of (i) $\theta = 0, \pi/2$ or (ii) $\omega \ll \omega_{ci}$ were just particular examples of the more general property that a cold plasma supports two distinct types of modes. Using a modest amount of algebraic manipulation (cf. assignments) it is straightforward to show that the quantity $B^2 - 4AC$ is positive definite for real θ, since

$$B^2 - 4AC = \left(S^2 - D^2 - SP\right)^2 \sin^4\theta + 4P^2D^2\cos^2\theta. \tag{6.50}$$

Thus, n is either pure real (corresponding to a propagating wave) or pure imaginary (corresponding to an evanescent wave).

From Eqs. (6.47) and (6.48) it is seen that cutoffs occur when $C = 0$, which happens if $P = 0$, $L = 0$, or $R = 0$. Also, resonances correspond to having $A \to 0$, in which case

$$S\sin^2\theta + P\cos^2\theta \simeq 0. \tag{6.51}$$

6.3.6 Wave normal surfaces

The information contained in a dispersion relation can be summarized in a qualitative, visual manner by a *wave normal surface*, which is a polar plot of the phase velocity of the wave normalized to c. Since $n = ck/\omega$, a wave normal surface is just a plot of $1/n(\theta)$ vs. θ. The most basic wave normal surface is obtained by considering the equation for a light wave in vacuum,

$$\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right)\mathbf{E} = 0, \tag{6.52}$$

which has the simple dispersion relation

$$\frac{1}{n^2} = \frac{\omega^2}{k^2c^2} = 1. \tag{6.53}$$

Thus, the wave normal surface of a light wave in vacuum is just a sphere of radius unity because $\omega/k = c/n$ is independent of direction. Wave normal surfaces of plasma waves are typically more complicated because n usually depends on θ.

The radius of the wave normal surface goes to zero at a resonance and goes to infinity at a cutoff (since $1/n \to 0$ at a resonance, $1/n \to \infty$ at a cutoff).

6.3.7 Taxonomy of modes – the CMA diagram

Equation (6.49) gives the general dispersion relation for arbitrary θ. While formally correct, this expression is of little practical value because of the complicated chain of dependence of n^2 on several variables. The CMA diagram (Clemmow and Mullaly (1955), Allis (1955)) provides an elegant method for revealing and classifying the large number of qualitatively different modes embedded in Eq. (6.49).

In principle, Eq. (6.49) gives the dependence of n^2 on the six parameters θ, ω, ω_{pe}, ω_{pi}, ω_{ce}, and ω_{ci}. However, ω_{pi} and ω_{pe} are not really independent parameters and neither are ω_{ci} and ω_{ce} because $\omega_{ce}^2/\omega_{ci}^2 = (m_i/m_e)^2$ and $\omega_{pe}^2/\omega_{pi}^2 = m_i/m_e$ for singly charged ions. Thus, once the ion species has been specified, the only free parameters are the density and the magnetic field. Once these have been specified, the plasma frequencies and the cyclotron frequencies are determined. It is reasonable to normalize these frequencies to the wave frequency in question since the quantities S, P, D depend only on the normalized frequencies. Thus, n^2 is effectively just a function of θ, ω_{pe}^2/ω^2, and ω_{ce}^2/ω^2. Pushing this simplification even further, we can say that for fixed ω_{pe}^2/ω^2 and ω_{ce}^2/ω^2, the refractive index n is just a function of θ. Then, once $n = n(\theta)$ is known, it can be used to plot a *wave normal surface*, i.e., ω/kc plotted vs. θ.

The CMA diagram is developed by first constructing a chart where the horizontal axis is $\ln\left(\omega_{pe}^2/\omega^2 + \omega_{pi}^2/\omega^2\right)$ and the vertical axis is $\ln\left(\omega_{ce}^2/\omega^2\right)$. For a given ω, any point on this chart corresponds to a unique density and a unique magnetic field. If we were ambitious, we could plot the wave normal surfaces $1/n$ vs. θ for a very large number of points on this chart, and so have plots of dispersions for a large set of cold plasmas. While conceivable, such a thorough examination of all possible combinations of density and magnetic field would require plotting an inconveniently large number of wave normal surfaces.

It is actually unnecessary to plot this very large set of wave normal surfaces because it turns out that the qualitative shape (i.e., topology) of the wave normal surfaces changes only at specific boundaries in parameter space. Away from these boundaries the wave normal surface deforms, but does not change its topology. The CMA diagram, shown in Fig. 6.2, charts these parameter space boundaries and so provides a powerful method for classifying cold plasma modes. Parameter space is divided up into a finite number of regions, called bounded volumes, separated by curves in parameter space, called bounding surfaces, across which the modes change *qualitatively*. Thus, within a bounded volume, modes change quantitatively but not qualitatively. For example, if Alfvén waves exist at one

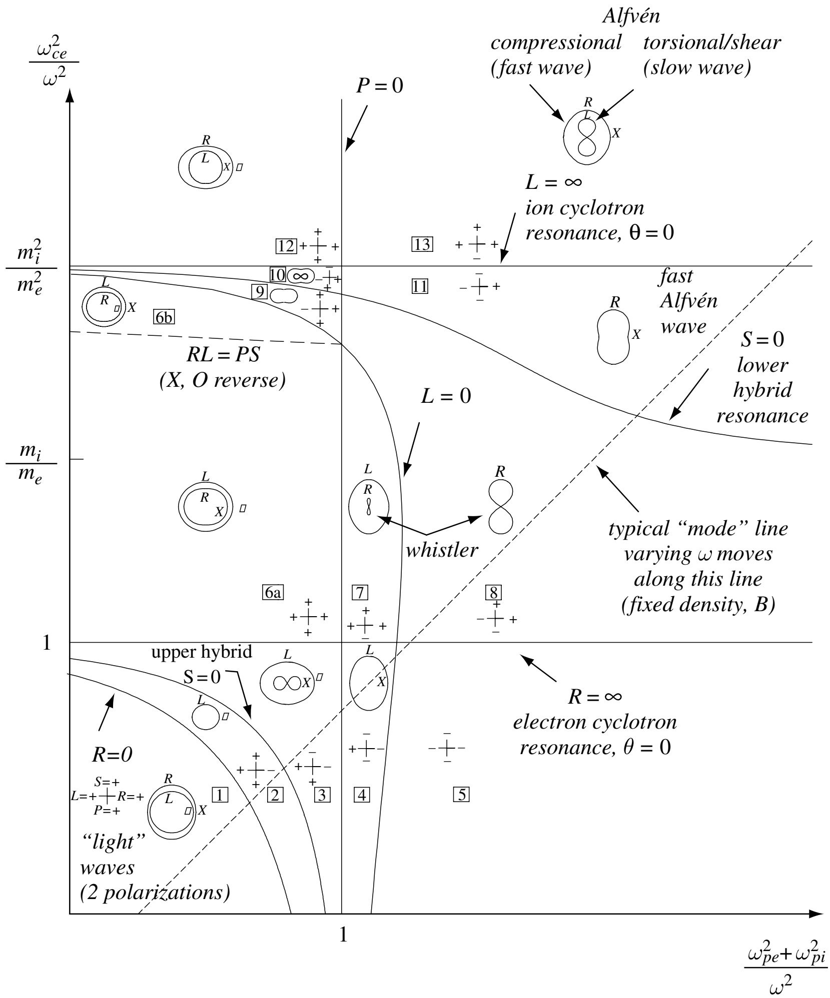

Fig. 6.2 CMA diagram.

point in a particular bounded volume, they must exist everywhere in that bounded volume, although the dispersion may not be quantitatively the same at different locations in the volume.

The appropriate choice of bounding surfaces consists of:

1. The *principle resonances*, which are the curves in parameter space where n^2 has a resonance at either $\theta = 0$ or $\theta = \pi/2$. Thus, the principal resonances are the curves $R = \infty$ (i.e., electron cyclotron resonance), $L = \infty$ (i.e., ion cyclotron resonance), and $S = 0$ (i.e., the upper and lower hybrid resonances).
2. The cutoffs $R = 0$, $L = 0$, and $P = 0$.

The behavior of wave normal surfaces inside a bounded volume and when crossing a bounded surface can be deduced using a set of five simple theorems (Stix 1962), each a consequence of the results derived so far:

1. Inside a bounded volume n cannot vanish. Proof: by setting $n = 0$ in Eq. (6.47) and using Eq. (6.48) it is seen that n vanishes only when $PRL = 0$, but $P = 0$, $R = 0$, and $L = 0$ have been defined to be bounding surfaces.
2. If n^2 has a resonance (i.e., goes to infinity) at *any* point in a bounded volume, then for *every* other point in the same bounded volume, there exists a resonance at some unique angle θ_{res} and its associated mirror angles, namely $-\theta_{res}$, $\pi - \theta_{res}$, and $-(\pi - \theta_{res})$, but at *no* other angles. Proof: if $n^2 \to \infty$ then $A \to 0$, in which case $\tan^2\theta_{res} = -P/S$ determines the unique θ_{res}. Now $\tan^2(\pi - \theta_{res}) = \tan^2\theta_{res}$ so there is also a resonance at the supplement $\theta = \pi - \theta_{res}$. Also, since the square of the tangent is involved, both θ_{res} and $\pi - \theta_{res}$ may be replaced by their negatives. Neither P nor S can change sign inside a bounded volume and both are single valued functions of their location in parameter space. Thus, $-P/S$ can only change sign at a bounding surface. In summary, if a resonance occurs at any point in a bounding surface, then a resonance exists at some unique angle θ_{res} and its associated mirror angles at every point in the bounding surface. Resonances only occur when P and S have opposite signs. Since $1/n$ goes to zero at a resonance, the radius of a wave normal surface goes to zero at a resonance.
3. At any point in parameter space and for a given interval in θ in which n is finite, n is either pure real or pure imaginary throughout that interval. Proof: n^2 is always real and is a continuous function of θ. The only situation where n can change from being pure real to being pure imaginary is when n^2 changes sign. This occurs when n^2 passes through zero but, because of the definition for bounding surfaces, n^2 does not vanish inside a bounded volume. Although n^2 may change sign when going through infinity this situation is not relevant because the theorem was restricted to finite n.
4. n is symmetric about $\theta = 0$ and $\theta = \pi/2$. Proof: n is a function of $\sin^2\theta$ and of $\cos^2\theta$, both of which are symmetric about $\theta = 0$ and $\theta = \pi/2$.
5. Except for the special case where the surfaces $PD = 0$ and $RL = PS$ intersect, the two modes may coincide only at $\theta = 0$ or at $\theta = \pi/2$. Proof: for $0 < \theta < \pi/2$ the square root in Eq. (6.49) is

$$\sqrt{B^2 - 4AC} = \sqrt{(RL - SP)^2 \sin^4\theta + 4P^2D^2\cos^2\theta} \tag{6.54}$$

   and can only vanish if $PD = 0$ and $RL = PS$ simultaneously.

These theorems provide sufficient information to characterize the morphology of wave normal surfaces throughout all of parameter space. In particular, the theorems show that only three types of wave normal surfaces exist. These are ellipsoid, dumbbell, and wheel as shown in Fig. 6.3(a)–(c) and each is a three-dimensional surface symmetric about the z axis.

We now discuss the features and interrelationships of these three types of wave normal surfaces. In this discussion, each of the two modes in Eq. (6.49)

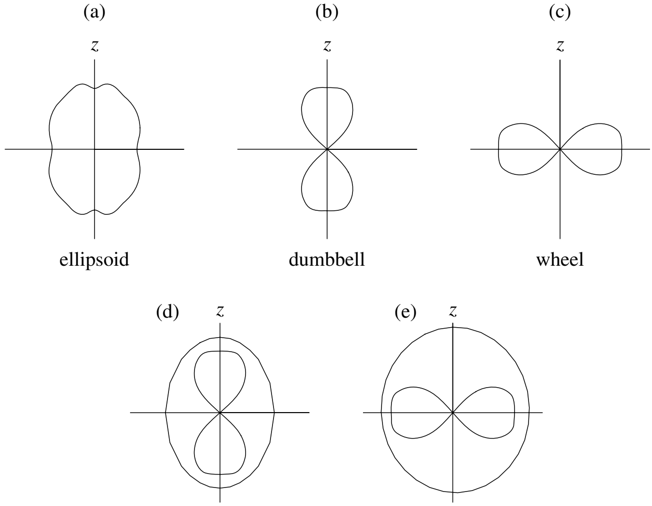

Fig. 6.3 (a), (b), (c) show types of wave normal surfaces; (d) and (e) show permissible overlays of wave normal surfaces.

is considered separately; i.e., either the plus or the minus sign is chosen. The convention is used that a mode is considered to exist (i.e., has a wave normal surface) only if $n^2 > 0$ for at least some range of θ; if $n^2 < 0$ for all angles, then the mode is evanescent (i.e., non-propagating) for all angles and is not plotted. The three types of wave normal surfaces are:

1. Bounded volume with no resonance and $n^2 > 0$ at some point in the bounded volume. Since $n^2 = 0$ occurs only at the bounding surfaces and $n^2 \to \infty$ only at resonances, n^2 must be positive and finite at every θ for each location in the bounded volume. The wave normal surface is thus ellipsoidal with symmetry about both $\theta = 0$ and $\theta = \pi/2$. The ellipse may deform as one moves inside the bounded volume, but will always have the morphology of an ellipse. This type of wave normal surface is shown in Fig. 6.3(a). The wave normal surface is three dimensional and is azimuthally symmetric about the z axis.
2. Bounded volume having a resonance at some angle θ_{res}, where $0 < \theta_{res} < \pi/2$ and $n^2(\theta)$ positive for $\theta < \theta_{res}$. At θ_{res}, $n \to \infty$ so the radius of the wave normal surface goes to zero. For $\theta < \theta_{res}$, the wave normal surface exists (n is pure real since $n^2 > 0$) and is plotted. At resonances $n^2(\theta)$ passes from $-\infty$ to $+\infty$ or vice versa. This type of wave normal surface is a dumbbell type as shown in Fig. 6.3(b).
3. Bounded volume having a resonance at some angle θ_{res}, where $0 < \theta_{res} < \pi/2$ and $n^2(\theta)$ positive for $\theta > \theta_{res}$. This is similar to case 2 above, except the wave normal surface now only exists for angles greater than θ_{res} and so has the wheel-type shape shown in Fig. 6.3(c).

Consider now the relationship between the two modes (plus and minus sign) given by Eq. (6.49). Because the two modes cannot intersect (cf. theorem 5 on p. 220) at angles other than $\theta = 0$, $\pi/2$, and mirror angles, if one mode is an ellipsoid and the other has a resonance (i.e., is a dumbbell or wheel), the ellipsoid must be outside the other dumbbell or wheel; for if not, the two modes would intersect at an angle other than $\theta = 0$ or $\theta = \pi/2$. This is shown in Figs. 6.3(d) and (e).

Also, only one of the modes can have a resonance, so at most one mode in a bounded volume can be a dumbbell or wheel. This can be seen by noting that a resonance occurs when $A \to 0$. In this case $B^2 \gg |4AC|$ in Eq. (6.49) and the two roots are *well separated*. This means the binomial expansion can be used on the square root in Eq. (6.49) to obtain

$$n^2 \simeq \frac{B \pm (B - 2AC/B)}{2A}$$

$$n_+^2 \simeq \frac{B}{A}, \quad n_-^2 \simeq \frac{C}{B}, \tag{6.55}$$

where $|n_+^2| \gg |n_-^2|$ since $B^2 \gg |4AC|$. The root n_+^2 has the resonance and the root n_-^2 has no resonance. Since the wave normal surface of the minus root has no resonance, its wave normal surface must be ellipsoidal (if it exists). Because the ellipsoidal surface must always lie outside the wheel or dumbbell surface, the ellipsoidal surface will have a larger value of ω/kc than the dumbbell or wheel at every θ and so the ellipsoidal mode will always be the *fast* mode. The mode with the resonance will be a dumbbell or wheel, will lie inside the ellipsoidal surface, and so will always be the *slow* mode. This concept of well-separated roots is quite useful and, if the roots are well separated, then Eq. (6.47) can be solved approximately for the large root (slow mode) by balancing the first two terms with each other, and for the small root (fast mode) by balancing the last two terms with each other.

Parameter space is subdivided into thirteen bounded volumes, each potentially containing two normal modes corresponding to two qualitatively distinct propagating waves. However, since two modes do not exist in all bounded volumes, the actual number of modes is smaller than 26. As an example of a bounded volume with two waves, the wave normal surfaces of the fast and slow Alfvén waves are in the upper right-hand corner of the CMA diagram, since this bounded volume corresponds to $\omega \ll \omega_{ci}$ and $\omega \ll \omega_{pe}$, i.e., above $L = 0$ and to the right of $P = 0$.

6.3.8 Use of the CMA diagram

The CMA diagram can be used in several ways. For example, it can be used to (i) identify all allowed cold plasma modes in a given plasma for various values

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of ω or (ii) investigate how a given mode evolves as it propagates through a spatially inhomogeneous plasma and possibly intersects resonances or cutoffs due to spatial variation of density or magnetic field.

Let us consider the first example. Suppose the plasma is uniform and has a prescribed density and magnetic field. Since $\ln\left[\left(\omega_{pe}^2+\omega_{pi}^2\right)/\omega^2\right]$ and $\ln\left[\omega_{ce}^2/\omega^2\right]$ are the coordinates of the CMA diagram, varying ω corresponds to tracing out a line having a slope of 45° and an offset determined by the prescribed density and magnetic field. High frequencies correspond to the lower left portion of this line and low frequencies to the upper right. Since the allowed modes lie along this line, if the line does not pass through a given bounded volume, then modes inside that bounded volume do not exist in the specified plasma.

Now consider the second example. Suppose the plasma is spatially non-uniform in such a way that both density and magnetic field are a function of position. To be specific, suppose density increases as one moves in the x direction while magnetic field increases as one moves in the y direction. Thus, the CMA diagram becomes a map of the actual plasma. A wave with prescribed frequency ω is launched at some position x, y and then propagates along some trajectory in parameter space as determined by its local dispersion relation. The wave will continuously change its character as determined by the local wave normal surface. Thus, a wave launched as a fast Alfvén mode from the upper-right bounded volume and propagating in the downward direction will only deform somewhat on traversing the $L=\infty$ bounding surface. In contrast, a wave launched as a slow Alfvén mode (dumbbell shape) from the same position will disappear when it reaches the $L=\infty$ bounding surface, because the slow mode does not exist on the lower side (high-frequency side) of the $L=\infty$ bounding surface. The slow Alfvén wave undergoes ion cyclotron resonance at the $L=\infty$ bounding surface and will be absorbed there.

6.4 Dispersion relation expressed as a relation between n_x^2 and n_z^2

The CMA diagram is very useful for classifying waves, but is often not so useful in practical situations where it is not obvious how to specify the angle θ. In a practical situation a wave is typically excited by an antenna that lies in a plane and the geometry of the antenna imposes the component of the wavevector in the antenna plane. The transmitter frequency determines ω.

For example, consider an antenna located in the $x=0$ plane and having some specified z dependence. When Fourier analyzed in z, such an antenna would excite a characteristic k_z spectrum. In the extreme situation of the antenna extending to infinity in the $x=0$ plane and having the periodic dependence $\exp(ik_z z)$, the antenna would excite just a single k_z. Thus, the antenna–transmitter combination in this situation would impose k_z and ω but leave k_x undetermined. The job of the dispersion relation would then be to determine k_x. It should be noted that antennas that are not both infinite and perfectly periodic will excite a spectrum of k_z modes rather than just a single k_z mode.

By writing $n_x^2 = n^2 \sin^2\theta$ and $n_z^2 = n^2\cos^2\theta$, Eqs. (6.47) and (6.48) can be expressed as a quadratic equation for n_x^2, namely

$$Sn_x^4 - [\bar{S}(S+P) - D^2]n_x^2 + P[\bar{S}^2 - D^2] = 0, \tag{6.56}$$

where

$$\bar{S} = S - n_z^2. \tag{6.57}$$

If the two roots of Eq. (6.56) are well separated, the large root is found by balancing the first two terms to obtain

$$n_x^2 \simeq \frac{\bar{S}(S+P) - D^2}{S}, \qquad \text{(large root)} \tag{6.58}$$

or in the limit of large P (i.e., low frequencies)

$$n_x^2 \simeq \frac{\bar{S}P}{S}, \qquad \text{(large root).} \tag{6.59}$$

The small root is found by balancing the last two terms of Eq. (6.56) to obtain

$$n_x^2 \simeq \frac{P[\bar{S}^2 - D^2]}{[\bar{S}(S+P) - D^2]}, \qquad \text{(small root)} \tag{6.60}$$

or in the limit of large P

$$n_x^2 \simeq \frac{(n_z^2 - R)(n_z^2 - L)}{S - n_z^2}, \qquad \text{(small root).} \tag{6.61}$$

Thus, any given n_z^2 always has an associated large n_x^2 mode and an associated small n_x^2 mode. Because the phase velocity is inversely proportional to the refractive index, the root with large n_x^2 is called the slow mode and the root with small n_x^2 is called the fast mode.

Using the quadratic formula it is seen that the exact form of these two roots of Eq. (6.56) is given by

$$n_x^2 = \frac{\overline{S}(S+P) - D^2 \pm \sqrt{\left[\overline{S}(S-P) - D^2\right]^2 + 4PD^2 n_z^2}}{2S}. \tag{6.62}$$

It is clear that n_x^2 can become infinite only when $S = 0$. Situations where n_x^2 is complex (i.e., neither pure real or pure imaginary) can occur when P is large and negative, in which case the argument of the square root can become negative. In these cases, θ also becomes complex and is no longer a physical angle. This shows that considering real angles between $0 < \theta < 2\pi$ does not account for all possible types of wave behavior. The regions where n_x^2 becomes complex is called a region of inaccessibility and is a region where Eq. (6.56) does not have real roots. If a plasma is non-uniform in the x direction so that S, P, and D are functions of x and $\omega^2 < \omega_{pe}^2 + \omega_{pi}^2$ so that P is negative, the boundaries of a region of inaccessibility (if such a region exists) are the locations where the square root in Eq. (6.62) vanishes, i.e., where there is a solution for $\overline{S}(S-P) - D^2 = \pm\sqrt{-4PD^2n_z^2}$.

6.5 A journey through parameter space

Imagine an enormous plasma where the density increases in the x direction and the magnetic field points in the z direction but increases in the y direction. Suppose further that a radio transmitter operating at a frequency ω is connected to a hypothetical antenna that emits *plane waves*, i.e., waves with spatial dependence $\exp(i\mathbf{k}\cdot\mathbf{x})$. These assumptions are somewhat self-contradictory because, in order to excite plane waves, an antenna must be infinitely long in the direction normal to $\mathbf{k}$ and if the antenna is infinitely long it cannot be localized. To circumvent this objection, it is assumed the plasma is so enormous that the antenna at any location is sufficiently large compared to the wavelength in question to emit waves that are nearly plane waves.

The antenna is located at some point x, y in the plasma and the emitted plane waves are detected by a phase-sensitive receiver. The position x, y corresponds to a point in CMA space. The antenna is rotated through a sequence of angles θ and, as the antenna is rotated, an observer walks in front of the antenna staying exactly one wavelength $\lambda = 2\pi/k$ from the face of the antenna. Since λ is proportional to $1/n = \omega/kc$ at fixed frequency, the locus of the observer's path will have the shape of a wave normal surface, i.e., a plot of $1/n$ versus θ.

Because of the way the CMA diagram was constructed, the topology of one of the two cold plasma modes always changes when a bounding surface is traversed. Which mode is affected and how its topology changes on crossing a bounding surface can be determined by monitoring the polarities of the four quantities S, P, R, L within each bounded volume. P changes polarity only at the $P = 0$ bounding surface, but R and L change polarity when they go through zero and also when they go through infinity. Furthermore, $S = (R+L)/2$ changes sign not only when $S = 0$ but also at $R = \infty$ and at $L = \infty$.

A straightforward way to establish how the polarities of S, P, R, L change as bounding surfaces are crossed is to start in the extreme lower left corner of parameter space, corresponding to $\omega^2 \gg \omega_{pe}^2, \omega_{ce}^2$. This is the limit of having no plasma and no magnetic field and so corresponds to unmagnetized vacuum. The cold plasma dispersion relation in this limit is simply $n^2 = 1$; i.e., vacuum electromagnetic waves such as ordinary light waves or radio waves. Here $S = P = R = L = 1$ because there are no plasma currents. Thus S, P, R, L are all *positive* in this bounded volume, denoted as Region 1 in Fig. 6.2 (regions are labeled by boxed numbers). To keep track of the respective polarities, a small cross is sketched in each of the 13 bounded volumes. The signs of L and R are noted on the left and right of the cross respectively, while the sign of S is shown at the top and the sign of P is shown at the bottom.

In traversing from Region 1 to Region 2, R passes through zero and so reverses polarity but the polarities of L, S, P are unaffected. Going from Region 2 to Region 3, S passes through zero so the sign of S reverses. By continuing from region to region in this manner, the plus or minus signs on the crosses in each bounded volume are established. It is important to remember that S changes sign at both $S = 0$ and the cyclotron resonances $L = \infty$ and $R = \infty$, but at all other bounding surfaces, only one quantity reverses sign.

Modes with resonances (i.e., dumbbells or wheels) only occur if S and P have opposite sign, which occurs in Regions 3, 7, 8, 10, and 13. The ordinary mode (i.e., $\theta = \pi/2$, $n^2 = P$) exists only if $P > 0$ and so exists only in regions to the left of the $P = 0$ bounding surface. Thus, to the right of the $P = 0$ bounding surface only extraordinary modes exist (i.e., only modes where $n^2 = RL/S$ at $\theta = \pi/2$). Extraordinary modes exist only if $RL/S > 0$, which cannot occur if an odd subset of the three quantities R, L, and S is negative. For example, in Region 5 all three quantities are negative so extraordinary modes do not exist in Region 5. The parallel modes $n^2 = R, L$ do not exist in Region 5 because R and L are negative there. Thus, no modes exist in Region 5 because if a mode were to exist there, it would need to have a limiting behavior of either ordinary or extraordinary at $\theta = \pi/2$ and of either right or left circularly polarized at $\theta = 0$.

When crossing a cutoff bounding surface ($R = 0, L = 0$, or $P = 0$), the outer (i.e., fast) mode has its wave normal surface become infinitely large, $\omega^2/k^2c^2 = 1/n^2 \to \infty$. Thus, immediately to the left of the $P = 0$ bounding surface, the fast mode (outer mode) is always the ordinary mode, because by definition this mode has the dispersion $n^2 = P$ at $\theta = \pi/2$ and so has a cutoff at $P = 0$. As one approaches the $P = 0$ bounding surface from the left, all the outer modes are ordinary modes and all disappear on crossing the $P = 0$ line so that to the right of the $P = 0$ line there are no ordinary modes.

In Region 13 where the modes are Alfvén waves, the slow mode is the $n^2 = L$ mode, since this is the mode that has the resonance at $L = \infty$. The slow Alfvén mode is the inertial Alfvén mode while the fast Alfvén mode is the compressional Alfvén mode. Going downwards from Region 13 to Region 11, the slow Alfvén wave undergoes ion cyclotron resonance and disappears, but the fast Alfvén wave remains. Similar arguments can be made to explain other boundary crossings in parameter space.

A subtle aspect of this taxonomy is the division of Region 6 into two subregions 6a, and 6b. This subtlety arises because the dispersion at $\theta = \pi/2$ has the form

$$n^2 = \frac{RL + PS \pm |RL - PS|}{2S} = P, RL/S. \tag{6.63}$$

In Region 6, both S and P are positive. If $RL - PS$ is also positive, then the plus sign gives the extraordinary mode, which is the slow mode (bigger n, inner of the two wave normal surfaces). On the other hand, if $RL - PS$ is negative, then the absolute value operator inverts the sign of $RL - PS$ and the minus sign now gives the extraordinary mode, which will be the fast mode (smaller n, outer of the two wave normal surfaces). Region 8 can also be divided into two regions (omitted here for clarity) separated by the $RL = PS$ line. In Region 8 the ordinary mode does not exist, but the extraordinary mode will be given by either the plus or minus sign in Eq. (6.63) depending on which side of the $RL = PS$ line one is considering.

For a given plasma density and magnetic field, varying the frequency corresponds to moving along a "mode" line that has a 45° slope on the log–log CMA diagram (see Fig. 6.2). If the plasma density is increased, the mode line moves to the right, whereas if the magnetic field is increased, the mode line moves up. Since any single mode line cannot pass through all 13 regions of parameter space, only a limited subset of the 13 regions of parameter space can be accessed for any given plasma density and magnetic field. Plasmas with $\omega_{pe}^2 > \omega_{ce}^2$ are often labeled "overdense" and plasmas with $\omega_{pe}^2 < \omega_{ce}^2$ are correspondingly labeled "underdense." For overdense plasmas, the mode line passes to the right of the intersection of the $P = 0$, $R = \infty$ bounding surfaces while for underdense plasmas the mode line passes to the left of this intersection. Two different plasmas will be self-similar if they have similar mode lines. For example, if a lab plasma has the same mode line as a space plasma it will support the same kind of modes, but do so in a scaled fashion. Because the CMA diagram is log–log the bounding surface curves extend infinitely to the left and right of the figure and also infinitely above and below it; however, no new regions exist outside of what is sketched in Fig. 6.2.

The weakly magnetized case corresponds to the lower parts of Regions 1–5, while the low-density case corresponds to the left parts of Regions 1, 2, 3, 6, 9, 10, and 12. The CMA diagram provides a visual way for categorizing a great deal of useful information. In particular, it allows identification of isomorphisms between modes in different regions of parameter space so that understanding developed about the behavior for one kind of mode can be exploited to explain the behavior of a different, but isomorphic, mode located in another region of parameter space.

6.6 High-frequency waves: Altar–Appleton–Hartree dispersion relation

Examination of the dielectric tensor elements S, P, and D shows that while both ion and electron terms are of importance for low-frequency waves, for high-frequency waves ($\omega \gg \omega_{ci}$, ω_{pi}) the ion terms are unimportant and may be dropped. Thus, for high-frequency waves the dielectric tensor elements simplify to

$$\begin{aligned} S &= 1 - \frac{\omega_{pe}^2}{\omega^2 - \omega_{ce}^2} \\ P &= 1 - \frac{\omega_{pe}^2}{\omega^2} \\ D &= \frac{\omega_{ce}}{\omega} \frac{\omega_{pe}^2}{\left(\omega^2 - \omega_{ce}^2\right)} \end{aligned} \tag{6.64}$$

and the corresponding R and L terms are

$$\begin{aligned} R &= 1 - \frac{\omega_{pe}^2}{\omega\left(\omega + \omega_{ce}\right)} \\ L &= 1 - \frac{\omega_{pe}^2}{\omega\left(\omega - \omega_{ce}\right)}. \end{aligned} \tag{6.65}$$

The development of long-distance short-wave radio communication in the 1930s motivated investigations into how radio waves bounce from the ionosphere. Because the bouncing involves a $P = 0$ cutoff and because the ionosphere has ω_{pe}^2 of order ω_{ce}^2 but usually larger, the relevant frequencies must be of the order of the electron plasma frequency and so are much higher than both the ion cyclotron and ion plasma frequencies. Thus, ion effects are unimportant and so all ion terms may be dropped in order to simplify the analysis.

Perhaps the most important result of this era was a peculiar, but useful, reformulation by Appleton, Hartree, and Altar[1] (Appleton 1932) of the $\omega \gg \omega_{pi}, \omega_{ci}$ limit of Eq. (6.49). The intent of this reformulation was to express n^2 in terms of its deviation from the vacuum limit, $n^2 = 1$. An obvious way to do this is to define $\xi = n^2 - 1$ and then rewrite Eq. (6.47) as an equation for ξ, namely

$$A(\xi^2 + 2\xi + 1) - B(\xi + 1) + C = 0 \tag{6.66}$$

or, after regrouping,

$$A\xi^2 + \xi(2A - B) + A - B + C = 0. \tag{6.67}$$

[1] See discussion by Swanson (1989) regarding the recent addition of Altar to this citation.

Unfortunately, when this expression is solved for ξ, the leading term is -1 and so this attempt to find the deviation of n^2 from its vacuum limit fails. However, a slight rewriting of Eq. (6.67) as

$$\frac{A-B+C}{\xi^2}+\frac{2A-B}{\xi}+A=0 \tag{6.68}$$

and then solving for $1/\xi$, gives

$$\xi=\frac{2(A-B+C)}{B-2A\pm\sqrt{B^2-4AC}}. \tag{6.69}$$

This expression does not have a leading term of -1 and so allows the solution of Eq. (6.49) to be expressed as

$$n^2=1+\frac{2(A-B+C)}{B-2A\pm\sqrt{B^2-4AC}}. \tag{6.70}$$

In the $\omega \gg \omega_{ci}, \omega_{pi}$ limit where S, P, D are given by Eq. (6.64), algebraic manipulation of Eq. (6.70) (cf. assignments) shows the numerator and denominator of the second term have a common factor. After canceling this common factor, Eq. (6.70) reduces to

$$n^2=1-\left[\frac{2\dfrac{\omega_{pe}^2}{\omega^2}\left(1-\dfrac{\omega_{pe}^2}{\omega^2}\right)}{2\left(1-\dfrac{\omega_{pe}^2}{\omega^2}\right)-\dfrac{\omega_{ce}^2}{\omega^2}\sin^2\theta\pm\Gamma}\right], \tag{6.71}$$

where

$$\Gamma=\sqrt{\frac{\omega_{ce}^4}{\omega^4}\sin^4\theta+4\frac{\omega_{ce}^2}{\omega^2}\left(1-\frac{\omega_{pe}^2}{\omega^2}\right)^2\cos^2\theta}. \tag{6.72}$$

Equation (6.71) is called the Altar–Appleton–Hartree dispersion relation (Appleton 1932) and has the desired property of showing the deviation of n^2 from the vacuum dispersion $n^2=1$.

We recall that the cold plasma dispersion relation simplified considerably when either $\theta=0$ or $\theta=\pi/2$. A glance at Eq. (6.71) shows that this expression reduces indeed to $n^2=R, L$ for $\theta=0$. Somewhat more involved manipulation shows that Eq. (6.71) also reduces to $n_+^2=P$ and $n_-^2=RL/S$ for $\theta=\pi/2$.

Equation (6.71) can be Taylor expanded in the vicinity of the principle angles $\theta=0$ and $\theta=\pi/2$ to give dispersion relations for quasi-parallel or quasi-perpendicular propagation. The terms quasi-longitudinal and quasi-transverse are commonly used to denote these situations. The nomenclature is somewhat unfortunate because of the possible confusion with the traditional convention that longitudinal and transverse refer to the orientation of **k** relative to the wave electric field. Here, longitudinal means **k** is nearly parallel to the static magnetic field while transverse means **k** is nearly perpendicular to the static magnetic field.

6.6.1 Quasi-transverse modes ($\theta \simeq \pi/2$)

For quasi-transverse propagation, the first term in Γ dominates, that is,

$$\frac{\omega_{ce}^4}{\omega^4}\sin^4\theta \gg 4\frac{\omega_{ce}^2}{\omega^2}\left(1-\frac{\omega_{pe}^2}{\omega^2}\right)^2\cos^2\theta. \tag{6.73}$$

In this case a binomial expansion of Γ gives

$$\begin{aligned}\Gamma &= \frac{\omega_{ce}^2}{\omega^2}\sin^2\theta\left[1+4\frac{\omega^2}{\omega_{ce}^2}\left(1-\frac{\omega_{pe}^2}{\omega^2}\right)^2\frac{\cos^2\theta}{\sin^4\theta}\right]^{1/2}\\ &\simeq \frac{\omega_{ce}^2}{\omega^2}\sin^2\theta+2\left(1-\frac{\omega_{pe}^2}{\omega^2}\right)^2\cot^2\theta.\end{aligned} \tag{6.74}$$

Substitution of Γ into Eq. (6.71) shows the generalization of the ordinary mode dispersion to angles in the vicinity of $\pi/2$ is

$$n_+^2 = \frac{1-\dfrac{\omega_{pe}^2}{\omega^2}}{1-\dfrac{\omega_{pe}^2}{\omega^2}\cos^2\theta}. \tag{6.75}$$

The subscript $+$ here means that the positive sign has been used in Eq. (6.71). This mode is called the QTO mode as an acronym for "quasi-transverse-ordinary."

Choosing the $-$ sign in Eq. (6.71) gives the quasi-transverse-extraordinary mode or QTX mode. After a modest amount of algebra (cf. assignments) the QTX dispersion is found to be

$$n_-^2 = \frac{\left(1-\dfrac{\omega_{pe}^2}{\omega^2}\right)^2-\dfrac{\omega_{ce}^2}{\omega^2}\sin^2\theta}{1-\dfrac{\omega_{pe}^2}{\omega^2}-\dfrac{\omega_{ce}^2}{\omega^2}\sin^2\theta}. \tag{6.76}$$

Note that the QTX mode has a resonance near the upper hybrid frequency.

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6.6.2 Quasi-longitudinal dispersion ($\theta \simeq 0$)

Here, the term containing $\cos^2\theta$ dominates in Eq. (6.72). Because there are no cancelations of the leading terms in Γ with any remaining terms in the denominator of Eq. (6.71), it suffices to keep only the leading term of Γ. Thus, in this limit

$$\Gamma \simeq 2\left|\frac{\omega_{ce}}{\omega}\left(1-\frac{\omega_{pe}^2}{\omega^2}\right)\cos\theta\right| = -2\left(1-\frac{\omega_{pe}^2}{\omega^2}\right)\left|\frac{\omega_{ce}}{\omega}\cos\theta\right|, \tag{6.77}$$

since $P = 1-\omega_{pe}^2/\omega^2$ is assumed to be negative. Upon substitution for Γ in Eq. (6.71) and then simplifying, one obtains

$$n_+^2 = 1-\frac{\omega_{pe}^2/\omega^2}{1-\left|\dfrac{\omega_{ce}}{\omega}\cos\theta\right|}, \quad \text{QLR mode} \tag{6.78}$$

and

$$n_-^2 = 1-\frac{\omega_{pe}^2/\omega^2}{1+\left|\dfrac{\omega_{ce}}{\omega}\cos\theta\right|}, \quad \text{QLL mode.} \tag{6.79}$$

These simplified dispersions are based on the implicit assumption that $|P|$ is large, because if $P \to 0$ the presumption that Eq. (6.77) gives the leading term in Γ would be inappropriate. When $\omega < |\omega_{ce}\cos\theta|$ the QLR mode (quasi-longitudinal, right-hand circularly polarized) is called the whistler or helicon wave. This wave is distinguished by having a descending whistling tone, which shows up at audio frequencies on sensitive amplifiers connected to long wire antennas. Whistlers may have been heard as early as the late nineteenth century by telephone linesmen installing long telephone lines. They became a subject of some interest in the trenches of the First World War when German scientist H. Barkhausen heard whistlers on a sensitive audio receiver while trying to eavesdrop on British military communications; the origin of these waves was a mystery at that time. After the war Barkhausen (1930) and Eckersley (1935) proposed that the descending tone was due to a dispersive propagation such that lower frequencies traveled more slowly, but did not explain the source location or propagation trajectory. The explanation had to wait over two more decades until Storey (1953) finally solved the mystery by showing that whistlers were caused by lightning bolts and identified two main types of propagation. The first type, called a short whistler, resulted from a lightning bolt in the opposite hemisphere exciting a wave that propagated dispersively along the Earth's magnetic field to the observer. The second type, called a long whistler, resulted from a lightning bolt in the vicinity of the observer exciting a wave that propagated dispersively along field lines to the opposite hemisphere, then reflected, and traveled back along the same path to the observer. The dispersion would be greater in this round-trip situation and also there would be a correlation with a click from the local lightning bolt. Whistlers are routinely observed by spacecraft flying through the Earth's magnetosphere and the magnetospheres of other planets.

The reason for the whistler's descending tone can be seen by representing each lightning bolt as a delta function in time

$$\delta(t) = \frac{1}{2\pi} \int \mathrm{e}^{-\mathrm{i}\omega t} \mathrm{d}\omega. \tag{6.80}$$

A lightning bolt therefore launches a very broad frequency spectrum. Because the ionospheric electron plasma frequency is in the range 10–30 MHz, audio frequencies are much lower than the electron plasma frequency, i.e., $\omega_{pe} \gg \omega$ and so $|P| \gg 1$. The electron cyclotron frequency in the ionosphere is of the order of 1 MHz so $\omega_{ce} \gg \omega$ also. Thus, the whistler dispersion for acoustic (a few kHz) waves in the ionosphere is

$$n_-^2 = \frac{\omega_{pe}^2}{\left| \dfrac{\omega_{ce}}{\omega} \cos\theta \right|} \tag{6.81}$$

or

$$k = \frac{\omega_{pe}}{c} \sqrt{\frac{\omega}{|\omega_{ce} \cos\theta|}}. \tag{6.82}$$

Each frequency ω in Eq. (6.80) has a corresponding k given by Eq. (6.82) so that the disturbance $g(x, t)$ excited by a lightning bolt has the form

$$g(x, t) = \frac{1}{2\pi} \int \mathrm{e}^{\mathrm{i}k(\omega)x - \mathrm{i}\omega t} \mathrm{d}\omega, \tag{6.83}$$

where $x = 0$ is the location of the lightning bolt. Because of the strong dependence of k on ω, contributions to the phase integral in Eq. (6.83) at adjacent frequencies will in general have substantially different phases. The integral can then be considered as the sum of contributions having all possible phases. Since there will be approximately equal amounts of positive and negative contributions, the contributions will cancel each other out when summed; this canceling is called phase mixing.

Suppose there exists some frequency ω at which the phase $k(\omega)x - \omega t$ has a local maximum or minimum with respect to variation of ω. In the vicinity of this extremum, the phase is independent of frequency and so the contributions from adjacent frequencies constructively interfere and produce a finite signal. Thus, an observer located at some position $x \neq 0$ will hear a signal only at the time when the phase in Eq. (6.83) is at an extremum. The phase extremum is found by setting to zero the derivative of the phase with respect to frequency, i.e., setting

$$\frac{\partial k}{\partial \omega} x - t = 0. \tag{6.84}$$

From Eq. (6.82) it is seen that

$$\frac{\partial k}{\partial \omega} = \frac{\omega_{pe}}{2c} \frac{1}{\sqrt{\omega |\omega_{ce} \cos\theta|}} \tag{6.85}$$

so the time at which a frequency ω is heard by an observer at location x is

$$t = \frac{\omega_{pe}}{2c} \frac{x}{\sqrt{\omega |\omega_{ce} \cos\theta|}}. \tag{6.86}$$

This shows that lower frequencies are heard at later times, resulting in the descending tone characteristic of whistlers.

6.7 Group velocity

Suppose that at time $t = 0$ the electric field of a particular fast or slow mode is decomposed into spatial Fourier modes, each varying as $\exp(\mathrm{i}\mathbf{k} \cdot x)$. The total wave field can then be written as

$$\mathbf{E}(\mathbf{x}) = \int \mathrm{d}\mathbf{k} \tilde{\mathbf{E}}(\mathbf{k}) \exp(\mathrm{i}\mathbf{k} \cdot \mathbf{x}), \tag{6.87}$$

where $\tilde{\mathbf{E}}(\mathbf{k})$ is the amplitude of the mode with wavevector $\mathbf{k}$. The dispersion relation assigns an ω to each $\mathbf{k}$, so that at later times the field evolves as

$$\mathbf{E}(\mathbf{x}, t) = \int \mathrm{d}\mathbf{k} \tilde{\mathbf{E}}(\mathbf{k}) \exp(\mathrm{i}\mathbf{k} \cdot \mathbf{x} - \mathrm{i}\omega(\mathbf{k})t), \tag{6.88}$$

where $\omega(\mathbf{k})$ is given by the dispersion relation. The integration over $\mathbf{k}$ may be viewed as a summation of rapidly oscillating waves, each having different rates of phase variation. In general, this sum vanishes because the waves add destructively or “phase mix.” However, if the waves add constructively, a finite $\mathbf{E}(\mathbf{x}, t)$ will result. Denoting the phase by

$$\phi(\mathbf{k}) = \mathbf{k} \cdot \mathbf{x} - \omega(\mathbf{k})t \tag{6.89}$$

it is seen that the Fourier components add constructively at extrema (minima or maxima) of $\phi(\mathbf{k})$ because, in the vicinity of an extremum, the phase is stationary with respect to $\mathbf{k}$, that is, the phase does not vary with $\mathbf{k}$. Thus, the trajectory $\mathbf{x} = \mathbf{x}(t)$, along which $\mathbf{E}(\mathbf{x}, t)$ is finite, is the trajectory along which the phase is stationary. At time t the stationary phase is the place where $\partial\phi(\mathbf{k})/\partial\mathbf{k}$ vanishes, which is where

$$\frac{\partial \phi}{\partial \mathbf{k}} = \mathbf{x} - \frac{\partial \omega}{\partial \mathbf{k}} t = 0. \tag{6.90}$$

The trajectory of the points of stationary phase is therefore

$$\mathbf{x}(t) = \mathbf{v}_g t, \tag{6.91}$$

where $\mathbf{v}_g = \partial\omega/\partial\mathbf{k}$ is called the group velocity. The group velocity is the velocity at which a pulse propagates in a dispersive medium and is also the velocity at which energy propagates.

The phase velocity for a one-dimensional system is defined as $v_{ph} = \omega/k$. In three dimensions this definition can be extended to be $\mathbf{v}_{ph} = \hat{k}\omega/k$, i.e., a vector in the direction of $\mathbf{k}$ but with the magnitude ω/k.

Group and phase velocities are the same only for the special case where ω is linearly proportional to k, a situation that occurs only if there is no plasma. For example, the phase velocity of electromagnetic plasma waves (dispersion $\omega^2 = \omega_{pe}^2 + k^2c^2$) is

$$\mathbf{v}_{ph} = \hat{k}\frac{\sqrt{\omega_{pe}^2 + k^2c^2}}{k}, \tag{6.92}$$

which is faster than the speed of light. However, no paradox results because information and energy travel at the group velocity, not the phase velocity. The group velocity for this wave is evaluated by taking the derivative of the dispersion with respect to $\mathbf{k}$ giving

$$2\omega\frac{\partial\omega}{\partial\mathbf{k}} = 2\mathbf{k}c, \tag{6.93}$$

or

$$\frac{\partial\omega}{\partial\mathbf{k}} = \frac{\mathbf{k}c}{\sqrt{\omega_{pe}^2 + k^2c^2}}, \tag{6.94}$$

which is less than the speed of light.

This illustrates an important property of the wave normal surface concept – a wave normal surface is a polar plot of the *phase velocity* and should not be confused with the group velocity.

6.8 Quasi-electrostatic cold plasma waves

Another useful way of categorizing waves is according to whether the wave electric field is:

1. electrostatic so that $\nabla\times\mathbf{E} = 0$ and $\mathbf{E} = -\nabla\phi$
   or
2. inductive so that $\nabla\cdot\mathbf{E} = 0$ and in Coulomb gauge, $\mathbf{E} = -\partial\mathbf{A}/\partial t$, where $\mathbf{A}$ is the vector potential.

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An electrostatic electric field is produced by net charge density whereas an inductive field is produced by time-dependent currents. Inductive electric fields are always associated with time-dependent magnetic fields via Faraday's law.

Waves involving purely electrostatic electric fields are called electrostatic waves, whereas waves involving inductive electric fields are called electromagnetic waves because these waves involve both electric and magnetic wave fields. In actuality, electrostatic waves must always have some slight inductive component, because there must always be a small current that establishes the net charge density. Thus, strictly speaking, the condition for electrostatic modes is $\nabla\times\mathbf{E}\simeq 0$ rather than $\nabla\times\mathbf{E}=0$.

In terms of Fourier modes where ∇ is replaced by $i\mathbf{k}$, electrostatic modes are those for which $\mathbf{k}\times\mathbf{E}=0$ so that $\mathbf{E}$ is parallel to $\mathbf{k}$; this means that electrostatic waves are longitudinal waves. Electromagnetic waves have $\mathbf{k}\cdot\mathbf{E}=0$ and so are transverse waves. Here, we are using the usual wave terminology where longitudinal and transverse refer to whether $\mathbf{k}$ is parallel or perpendicular to $\mathbf{E}$.

The electron plasma waves and ion acoustic waves discussed in the previous chapter were electrostatic, the compressional Alfvén wave was inductive, and the inertial Alfvén wave was both electrostatic and inductive. We now wish to show that in a *magnetized* plasma, the wheel and dumbbell modes in the CMA diagram always have electrostatic behavior in the region where the wave normal surface comes close to the origin, i.e., near the cross-over in the figure-eight pattern of these wave normal surfaces. For these waves, when n becomes large (i.e., near the cross-over of the figure-eight pattern of the wheel or dumbbell), $\mathbf{n}$ becomes nearly parallel to $\mathbf{E}$ and the magnetic part of the wave becomes infinitesimal. We now prove this assertion and also take care to distinguish this situation from another situation where n becomes infinite, namely at cyclotron resonances.

When the two roots of the dispersion $An^4-Bn^2+C=0$ are well separated (i.e., $B^2\gg 4AC$) the slow mode is found by assuming that n^2 is large. In this case the dispersion can be approximated as $An^4-Bn^2\simeq 0$, which gives the slow mode as $n^2\simeq B/A$. Resonance (i.e., $n^2\to\infty$) can thus occur either from

1. $A=S\sin^2\theta+P\cos^2\theta$ vanishing, or
2. $B=RL\sin^2\theta+PS(1+\cos^2\theta)$ becoming infinite.

These two cases are different. In the first case S and P remain finite and the vanishing of A determines a critical angle $\theta_{res}=\tan^{-1}\sqrt{-P/S}$; this angle is the cross-over angle of the figure-eight pattern of the wheel or dumbbell. In the second case either R or L must become infinite, a situation occurring only at the R or L bounding surfaces, i.e., only at the electron or ion cyclotron resonances.

The first case results in quasi-electrostatic cold plasma waves, whereas the second case does not. To see this, the electric field is first decomposed into its longitudinal and transverse parts

$$\begin{aligned}\mathbf{E}^l &= \hat{n}\hat{n}\cdot\mathbf{E}\\ \mathbf{E}^t &= \mathbf{E}-\mathbf{E}^l,\end{aligned} \tag{6.95}$$

where $\hat{n}=\hat{k}=\mathbf{n}/n$ is a unit vector in the direction of $\mathbf{n}$. The cold plasma wave equation, Eq. (6.17), can thus be written as

$$\mathbf{nn}\cdot\left(\mathbf{E}^t+\mathbf{E}^l\right)-n^2\left(\mathbf{E}^t+\mathbf{E}^l\right)+\overleftrightarrow{\mathbf{K}}\cdot\left(\mathbf{E}^t+\mathbf{E}^l\right)=0. \tag{6.96}$$

Since $\mathbf{n}\cdot\mathbf{E}^t=0$ and $\mathbf{nn}\cdot\mathbf{E}^l=n^2\mathbf{E}^l$, this expression can be recast as

$$\left(\overleftrightarrow{\mathbf{K}}-n^2\overleftrightarrow{\mathbf{I}}\right)\cdot\mathbf{E}^t+\overleftrightarrow{\mathbf{K}}\cdot\mathbf{E}^l=0, \tag{6.97}$$

where $\overleftrightarrow{\mathbf{I}}$ is the unit tensor. If the resonance is such that

$$n^2 \gg K_{ij}, \tag{6.98}$$

where K_{ij} are the elements of the dielectric tensor, then Eq. (6.97) may be approximated as

$$-n^2\mathbf{E}^t+\overleftrightarrow{\mathbf{K}}\cdot\mathbf{E}^l\simeq 0. \tag{6.99}$$

This shows that the transverse electric field is

$$\mathbf{E}^t=\frac{1}{n^2}\overleftrightarrow{\mathbf{K}}\cdot\mathbf{E}^l, \tag{6.100}$$

which is much smaller in magnitude than the longitudinal electric field by virtue of Eq. (6.98).

An easy way to obtain the dispersion relation (determinant of this system) is to dot Eq. (6.100) with $\mathbf{n}$ to obtain

$$\mathbf{n}\cdot\overleftrightarrow{\mathbf{K}}\cdot\mathbf{n}=n^2(S\sin^2\theta+P\cos^2\theta)\simeq 0, \tag{6.101}$$

which is just the first case discussed above. This argument is self-consistent because for the first case (i.e., $A\to 0$) the quantities S, P, D remain finite so the condition given by Eq. (6.98) is satisfied.

The second case, $B\to\infty$, occurs at the cyclotron resonances where S and D diverge so the condition given by Eq. (6.98) is not satisfied. Thus, for the second case the electric field is not quasi-electrostatic.

6.9 Resonance cones

The situation $A\to 0$ corresponds to Eq. (6.101) which is a dispersion relation having the surprising property of depending on θ, but *not* on the magnitude of n.

This limiting form of dispersion has some bizarre aspects, which will now be examined.

The group velocity in this situation can be evaluated by writing Eq. (6.101) as

$$k_x^2 S + k_z^2 P = 0 \tag{6.102}$$

and then taking the vector derivative with respect to $\mathbf{k}$ to obtain

$$2k_x \hat{x} S + 2k_z \hat{z} P + \left(k_x^2 \frac{\partial S}{\partial \omega} + k_z^2 \frac{\partial P}{\partial \omega} \right) \frac{\partial \omega}{\partial \mathbf{k}} = 0, \tag{6.103}$$

which may be solved to give

$$\frac{\partial \omega}{\partial \mathbf{k}} = -2 \left(\frac{k_x \hat{x} S + k_z \hat{z} P}{k_x^2 \dfrac{\partial S}{\partial \omega} + k_z^2 \dfrac{\partial P}{\partial \omega}} \right). \tag{6.104}$$

If Eq. (6.104) is dotted with $\mathbf{k}$ the surprising result

$$\mathbf{k} \cdot \frac{\partial \omega}{\partial \mathbf{k}} = 0 \tag{6.105}$$

is obtained, which means that the group velocity is *orthogonal* to the phase velocity. The same result may also be obtained in a quicker but more abstract way by using spherical coordinates in k-space, in which case the group velocity is just

$$\frac{\partial \omega}{\partial \mathbf{k}} = \hat{\mathbf{k}} \frac{\partial \omega}{\partial k} + \frac{\hat{\theta}}{k} \frac{\partial \omega}{\partial \theta}. \tag{6.106}$$

Applying this to Eq. (6.101), it is seen that the first term on the right-hand side vanishes because the dispersion relation is independent of the magnitude of $\mathbf{k}$. Thus, the group velocity is in the $\hat{\theta}$ direction, so the group and phase velocities are again *orthogonal*, since $\mathbf{k}$ is orthogonal to $\hat{\theta}$. Thus, energy and information propagate at right angles to the phase velocity.

A more physically intuitive interpretation of this phenomenon may be developed by "un-Fourier" analyzing the cold plasma wave equation, Eq. (6.17), giving

$$\nabla \times \nabla \times \mathbf{E} - \frac{\omega^2}{c^2} \overleftrightarrow{\mathbf{K}} \cdot \mathbf{E} = 0. \tag{6.107}$$

The modes corresponding to $A \to 0$ were obtained by dotting the dispersion relation with $\mathbf{n}$, an operation equivalent to taking the divergence in real space, and then arguing that the wave is mainly longitudinal. Let us therefore assume that $\mathbf{E} \simeq -\nabla \phi$ and take the divergence of Eq. (6.107) to obtain

$$\nabla \cdot \left(\overleftrightarrow{\mathbf{K}} \cdot \nabla \phi \right) = 0, \tag{6.108}$$

which is essentially Poisson's equation for a medium having dielectric tensor $\overleftrightarrow{\mathbf{K}}$. Equation (6.108) can be expanded to give

$$S\frac{\partial^2\phi}{\partial x^2}+P\frac{\partial^2\phi}{\partial z^2}=0. \tag{6.109}$$

If S and P have the same sign, Eq. (6.109) is an elliptic partial differential equation and so is just a distorted form of Poisson's equation. In fact, by defining the stretched coordinates $\xi=x/\sqrt{|S|}$ and $\eta=z/\sqrt{|P|}$, Eq. (6.109) becomes Poisson's equation in $\xi-\eta$ space.

Suppose now that waves are being excited by a line source $q\delta(x)\delta(z)\exp(-\mathrm{i}\omega t)$, i.e., a wire antenna lying along the y axis oscillating at the frequency ω. In this case Eq. (6.109) becomes

$$\frac{\partial^2\phi}{\partial\xi^2}+\frac{\partial^2\phi}{\partial\eta^2}=\frac{q}{|SP|^{3/2}\varepsilon_0}\delta(\xi)\delta(\eta) \tag{6.110}$$

so that the equipotential contours excited by the line source are just static concentric circles in ξ, η or, equivalently, static concentric ellipses in x, z.

However, if S and P have *opposite signs*, the situation is entirely different, because now the equation is hyperbolic and has the form

$$\frac{\partial^2\phi}{\partial x^2}=\left|\frac{P}{S}\right|\frac{\partial^2\phi}{\partial z^2}. \tag{6.111}$$

Equation (6.111) is formally analogous to the standard hyperbolic wave equation

$$\frac{\partial^2\psi}{\partial t^2}=c^2\frac{\partial^2\psi}{\partial z^2}, \tag{6.112}$$

which has solutions propagating along the characteristics $\psi=\psi(z\pm ct)$. Thus, the solutions of Eq. (6.111) also propagate along characteristics, i.e.,

$$\phi=\phi\left(z\pm\sqrt{-P/S}x\right), \tag{6.113}$$

which are characteristics in the $x-z$ plane rather than the $x-t$ plane. For a line source, the potential is infinite at the line, and this infinite potential propagates from the source following the characteristics

$$z=\pm\sqrt{-\frac{P}{S}}x. \tag{6.114}$$

If the source is a point source, then the potential has the form

$$\phi(r,z)\sim\frac{q}{4\pi\varepsilon_0\left(\dfrac{r^2}{S}+\dfrac{z^2}{P}\right)^{1/2}}, \tag{6.115}$$

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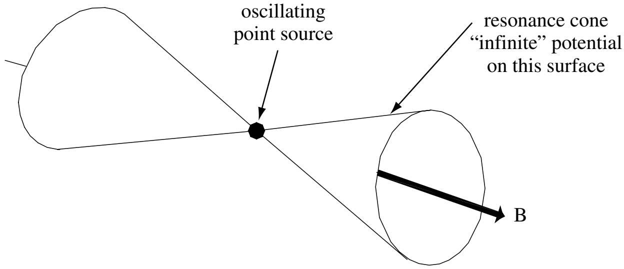

Fig. 6.4 Resonance cone excited by oscillating point source in magnetized plasma.

which diverges on the conical surface having cone angle $\tan\theta_{cone} = r/z = \pm\sqrt{-S/P}$, as shown in Fig. 6.4. These singular surfaces are called resonance cones and were first observed by Fisher and Gould (1969). The singularity results because the cold plasma approximation allows k to be arbitrarily large (i.e., allows infinitesimally short wavelengths). However, when k is made larger than ω/v_T, the cold plasma assumption $\omega/k \gg v_T$ becomes violated and warm plasma effects need to be taken into account. Thus, instead of becoming infinite on the resonance cone, the potential is large and finite and has a fine structure determined by thermal effects (Fisher and Gould 1971).

Resonance cones exist in the following regions of parameter space:

(i) Region 3, where they are called upper hybrid resonance cones; here $S < 0, P > 0$.

(ii) Regions 7 and 8, where they are a limiting form of the whistler wave and are also called lower hybrid resonance cones, since they are affected by the lower hybrid resonance (Briggs and Parker 1972, Bellan and Porkolab 1975). For $\omega_{ci} \ll \omega \ll \omega_{pe}, \omega_{ce}$ the P and S dielectric tensor elements become

$$P \simeq -\frac{\omega_{pe}^2}{\omega^2}, \quad S \simeq 1 - \frac{\omega_{pi}^2}{\omega^2} + \frac{\omega_{pe}^2}{\omega_{ce}^2}, \tag{6.116}$$

so the cone angle $\theta_{cone} = \tan^{-1} r/z$ is

$$\theta_{cone} = \tan^{-1}\sqrt{-\frac{S}{P}} = \tan^{-1}\sqrt{\left(\omega^2 - \omega_{lh}^2\right)\left(\omega_{pe}^{-2} + \omega_{ce}^{-2}\right)}. \tag{6.117}$$

If $\omega \gg \omega_{lh}$, the cone depends mainly on the smaller of ω_{pe}, ω_{ce}. For example, if $\omega_{ce} \ll \omega_{pe}$ then the cone angle is simply

$$\theta_{cone} \simeq \tan^{-1}\omega/\omega_{ce}, \tag{6.118}$$

whereas if $\omega_{ce} \gg \omega_{pe}$ then

$$\theta_{cone} \simeq \tan^{-1}\omega/\omega_{pe}. \tag{6.119}$$

For low-density plasmas this last expression can be used as the basis for a simple, accurate plasma density diagnostic.

(iii) Regions 10 and 13. The Alfvén resonance cones in Region 13 have a cone angle $\theta_{cone} = \omega/\sqrt{|\omega_{ce}\omega_{ci}|}$ and are associated with the electrostatic limit of inertial Alfvén waves (Stasiewicz *et al.* 2000). To the best of the author's knowledge, cones have not been investigated in Region 10, which corresponds to an unusual mix of parameters, namely ω_{pe} is the same order of magnitude as ω_{ci}.

6.10 Assignments

1. Prove that the cold plasma dispersion relation can be written as

$$An^4 - Bn^2 + C = 0,$$

where

$$\begin{aligned} A &= S\sin^2\theta + P\cos^2\theta \\ B &= (S^2 - D^2)\sin^2\theta + SP(1+\cos^2\theta) \\ C &= P(S^2 - D^2) \end{aligned}$$

so that the dispersion is

$$n^2 = \frac{B \pm \sqrt{B^2 - 4AC}}{2A}.$$

Prove that

$$RL = S^2 - D^2.$$

2. Prove that n^2 is always real if θ is real, by showing

$$B^2 - 4AC = \left[S^2 - D^2 - SP\right]^2 \sin^4\theta + 4P^2D^2\cos^2\theta.$$

3. Plot the bounding surfaces of the CMA diagram, by defining $m_e/m_i = \lambda$, $x = \left(\omega_{pe}^2 + \omega_{pi}^2\right)/\omega^2$, and $y = \omega_{ce}^2/\omega^2$. Show that

$$\frac{\omega_{pe}^2}{\omega^2} = \frac{x}{1+\lambda}$$

so

$$P = 1 - x$$

and S, R, L are functions of x and y with λ as a parameter. Hint: it is easier to plot x versus y for some of the functions.
4. Plot n^2 versus ω for $\theta = \pi/2$, showing the hybrid resonances.
5. Starting in Region 1 of the CMA diagram, establish the signs of S, P, R, L in all the regions.
6. Plot the CMA mode lines for plasmas having $\omega_{pe}^2 \gg \omega_{ce}^2$ and vice versa.

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7. Consider a plasma with two ion species. By plotting S versus ω show there is an ion–ion hybrid resonance between the two ion cyclotron frequencies (Buchsbaum 1960, Ono 1979). Give an approximate expression for the frequency of this resonance in terms of the ratios of the densities of the two ion species. Hint: compare the magnitude of the electron term to that of the two ion terms. Using quasi-neutrality, obtain an expression depending only on the fractional density of each ion species.
8. Consider a two-dimensional plasma with an oscillatory delta function source at the origin. Suppose that slow waves are being excited, which satisfy the electrostatic dispersion

$$k_x^2 S + k_z^2 P = 0,$$

where $SP < 0$. By writing the source on the z axis as

$$f(z) = \frac{1}{2\pi} \int \mathrm{e}^{\mathrm{i}k_z z - \mathrm{i}\omega t} \mathrm{d}k_z$$

and by solving the dispersion to give $k_x = k_x(k_z)$ show the potential excited in the plasma is singular along the resonance cone surfaces. Explain why this happens. Draw the group and phase velocity directions.
9. What is the polarization (i.e., relative magnitude of E_x, E_y, E_z) of the QTO, QTX, and the two QL modes? How should a microwave horn be oriented (i.e., in which way should the **E** field of the horn point) when being used for (i) a QTO experiment, (ii) a QTX experiment? Which experiment would be best suited for heating the plasma and which best suited for measuring the density of the plasma?
10. Show there is a simple factoring of the cold plasma dispersion relation in the low frequency limit $\omega \ll \omega_{ci}$. Hint: first find approximate forms of S, P, and D in this limit and then show that the cold plasma dielectric tensor becomes diagonal. Consider a mode that has only E_y finite and a mode that only has E_x and E_z finite. What are the dispersion relations for these two modes, expressed in terms of ω as a function of k? Assume the Alfvén velocity is much smaller than the speed of light.